

PVLoad-R1 Bench Card

WIRING THE ARDUINO

Connect ten wires from the Arduino to the 10-pin header on the board. Pin 1 is the square pad.

1	5V	5 V
2	GND	GND
3	D6	K1 drive
4	D7	K2 drive
5	D8	K3 drive
6	D9	CS U2
7	D10	CS U1
8	D11	SDI
9	D12	SDO
10	D13	SCK

On the Uno, 5V and GND are on the power side. D8 to D13 are the six pins nearest the USB socket.

THE THREE BANANA JACKS

J1, red. PV+, the cell's positive.

J2, red. +24 V from the bench supply.

J3, black. Ground. Connect both the supply negative and the cell negative here. Use a stacking plug. Do not force two plugs into one jack.

POWER UP

1. 24 V on
2. 5 V on (plug in the Arduino)
3. PV− → J3
4. PV+ → J1
5. illuminate the cell

POWER DOWN

1. darken the cell
2. PV+ out of J1
3. PV− out of J3
4. Arduino out (5 V off)
5. 24 V off

WHAT ON AND OFF MEAN

Do not unplug anything from the board to turn a supply on or off. The supply leads stay in J2 and J3. The ten header wires stay on the Arduino.

For 24 V, press the output button on the bench supply. If the supply has no output button, put the black lead in J3 first, then push the red lead into J2.

For 5 V, plug or unplug the USB cable at the laptop end. Stop the script first.

USING THE HANDHELD METER

Put the black lead in the socket marked **COM**. Put the red lead in the socket marked **VΩ**. Never use the **10A** socket. That socket makes the meter a piece of wire and shorts J1 to J3.

Turn the dial to **ohms** (Ω) to read the board's resistance across J1 and J3. Turn it to **DC volts** (V with a straight line over dashes) to probe pins against ground.

CABLES TO THE LAPTOP

Arduino. A standard USB printer cable, A to B.

EDFA100P laser. A standard USB cable.

Bench supply. Two banana leads to J2 and J3. Set 24.0 V and a current limit near 100 mA.

The two Keithley 617 meters. These have no USB. They have one GPIB socket, the wide connector with two screws. Use a single USB to GPIB adapter, National Instruments brand. Plug the adapter into the laptop. Run a GPIB cable from the adapter to the first meter. GPIB connectors stack, so run a second GPIB cable from the first meter to the second. Both meters share the one adapter.

Set a different address on each meter before you connect them. Both meters arrive on address 27, and two meters cannot share one address. Set the volts meter to **22** and the amps meter to **23**.

CONNECTING THE 617 METERS TO THE CELL

The front socket marked INPUT is triaxial, so a banana plug does not fit. Screw on a triax to dual banana adapter. This gives you two banana posts, **HI** and **LO**.

The amps meter goes inside the wire. Current must flow through the meter. Cut the wire from the cell's + terminal to J1 and put the meter in the break.

```
cell +   → HI post
LO post  → J1
```

The volts meter goes across the cell. Cut nothing. Two leads touch the cell's terminals.

```
HI post  → cell +
LO post  → cell −
```

Clip these to the cell's own leads, not to J1 and J3. On the jacks the reading passes through the amps meter and comes out low.

USING THE 617 METERS

The power switch is on the front.

At power up the meter is set to Volts and **ZERO CHECK** is on. With ZERO CHECK on, the meter reads zero whatever you connect. Press ZERO CHECK to turn it off before you measure by hand. The script turns it off itself when it drives the meter.

The function buttons select V, A, ohms or coulombs. The script sets these too.

To set the GPIB address: press SELECT until IEEE appears. Set the number with the ADJUST arrows. Press SHIFT, then SELECT, to store and exit.

REMOTE lights while the laptop drives the meter. Press LOCAL to take manual control back.

THE LASER INTERLOCK

A small connector on the back of the amplifier. It is a safety cutout for a door switch. Plug in the shorting plug supplied with the unit. If the connector is empty, the laser does not fire.

LASER SAFETY. Class 3B, 1550 nm. You cannot see this light. The amplifier emits up to 30 mW whenever it is enabled, at any pump current, and with no optical input. Terminate the output. Never look into the fibre.

DO NOT. Never leave the cell connected and lit while 24 V is off. If 5 V is on and 24 V is off, the cell's current flows into the chip and destroys it. Connect the cell last and disconnect it first.